Q1. JEE Main 2026 (23 January Shift 1)

Let the domain of the function $f(x)=\log_3 \log_5 \log_7\left(9x-x^2-13\right)$ be the interval (m, n). Let the hyperbola $\frac{x^2}{\mathrm{a}^2}-\frac{y^2}{\mathrm{b}^2}=1$ have eccentricity $\frac{\mathrm{n}}{3}$ and the length of the latus rectum $\frac{8\,\mathrm{m}}{3}$. Then $\mathrm{b}^2-\mathrm{a}^2$ is equal to :

(1) 7 (2) 11 (3) 5 (4) 9

Q2. JEE Main 2026 (22 January Shift 2)

Let $\mathrm{P}(10, 2\sqrt{15})$ be a point on the hyperbola $\frac{x^2}{\mathrm{a}^2}-\frac{y^2}{\mathrm{b}^2}=1$, whose foci are S and S′. If the length of its latus rectum is 8, then the square of the area of $\Delta \mathrm{PSS}'$ is equal to :

(1) 900 (2) 4200 (3) 1462 (4) 2700

Q3. JEE Main 2026 (22 January Shift 1)

If the line $\alpha x+2y=1$, where $\alpha \in \mathbb{R}$, does not meet the hyperbola $x^2-9y^2=9$, then a possible value of α is:

(1) 0.5 (2) 0.6 (3) 0.7 (4) 0.8

Q4. JEE Main 2026 (23 January Shift 2)

Let PQ be a chord of the hyperbola $\frac{x^2}{4}-\frac{y^2}{b^2}=1$, perpendicular to the x -axis such that OPQ is an equilateral triangle, O being the centre of the hyperbola. If the eccentricity of the hyperbola is $\sqrt{3}$, then the area of the triangle OPQ is

(1) $2\sqrt{3}$ (2) $\frac{11}{5}$ (3) $\frac{9}{5}$ (4) $\frac{8\sqrt{3}}{5}$

Q5. JEE Main 2026 (21 January Shift 1)

Let the foci of a hyperbola coincide with the foci of the ellipse $\frac{x^2}{36}+\frac{y^2}{16}=1$. If the eccentricity of the hyperbola is 5, then the length of its latus rectum is :

(1) 16 (2) $24\sqrt{5}$ (3) 12 (4) $\frac{96}{\sqrt{5}}$

Q6. JEE Main 2026 (28 January Shift 1)

For some $\theta \in \left(0, \frac{\pi}{2}\right)$, let the eccentricity and the length of the latus rectum of the hyperbola $x^2-y^2 \sec^2\theta=8$ be e_1 and l_1, respectively, and let the eccentricity and the length of the latus rectum of the ellipse $x^2 \sec^2\theta+y^2=6$ be e_2 and l_2, respectively. If $e_1^2=e_2^2\left(\sec^2\theta+1\right)$, then $\left(\frac{l_1 l_2}{e_1 e_2}\right)\tan^2\theta$ is equal to ____

ANSWERS AND SOLUTIONS

1. (1) **2.** (4) **3.** (4) **4.** (4) **5.** (4) **6.** 8

1. **(1)** For $f(x) = \log_3 \log_5 \log_7(9x - x^2 - 13)$:
$\log_7(\cdot) > 0 \Rightarrow 9x - x^2 - 13 > 1 \Rightarrow (x-2)(x-7) < 0 \Rightarrow x \in (2, 7)$.
$\log_5(\cdot) > 0 \Rightarrow \log_7(9x - x^2 - 13) > 1 \Rightarrow 9x - x^2 - 13 > 7 \Rightarrow (x-4)(x-5) < 0 \Rightarrow x \in (4, 5)$.
Domain $= (4, 5)$, so $m = 4, n = 5$.
Eccentricity $e = n/3 = 5/3$, latus rectum $= 8m/3 = 32/3$.
$e^2 = 1 + b^2/a^2 \Rightarrow b^2/a^2 = 16/9$.
$2b^2/a = 32/3 \Rightarrow 32a/9 = 32/3 \Rightarrow a = 3$, so $a^2 = 9$, $b^2 = 16$.
$b^2 - a^2 = 16 - 9 = 7$.

2. **(4)** For the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, point $P(10, 2\sqrt{15})$ gives $\frac{100}{a^2} - \frac{60}{b^2} = 1$.
The latus rectum length $\frac{2b^2}{a} = 8$ gives $b^2 = 4a$.
Substituting: $\frac{100}{a^2} - \frac{15}{a} = 1$, which yields $a^2 + 15a - 100 = 0$, so $a = 5$ and $b^2 = 20$.
Thus $c^2 = 45$ and $c = 3\sqrt{5}$.
The foci are $S(\pm 3\sqrt{5}, 0)$.
Triangle PSS' has base $SS' = 6\sqrt{5}$ and height $2\sqrt{15}$.
Area $= \frac{1}{2} \cdot 6\sqrt{5} \cdot 2\sqrt{15} = 30\sqrt{3}$.
Therefore $(\text{Area})^2 = 2700$.

3. **(4)** Hyperbola: $\frac{x^2}{9} - y^2 = 1$. Line: $y = \frac{1-\alpha x}{2}$.
Substituting: $x^2 - 9 \cdot \frac{(1-\alpha x)^2}{4} = 9$
$\Rightarrow (4 - 9\alpha^2)x^2 + 18\alpha x - 45 = 0$.
For no intersection (when $4 - 9\alpha^2 \neq 0$): discriminant < 0.
$D = 324\alpha^2 + 180(4 - 9\alpha^2) < 0$
$\Rightarrow -1296\alpha^2 + 720 < 0 \Rightarrow \alpha^2 > \frac{5}{9}$.
$|\alpha| > \frac{\sqrt{5}}{3} \approx 0.745$.
Among the options, only $\alpha = 0.8$ satisfies this condition.

4. **(4)** Hyperbola $\frac{x^2}{4} - \frac{y^2}{b^2} = 1$ with $e = \sqrt{3}$. $e^2 = 1 + \frac{b^2}{4} = 3 \Rightarrow b^2 = 8$.
Let $P = (x_0, y_0)$, $Q = (x_0, -y_0)$.
For equilateral $\triangle OPQ$:
$OP = PQ \Rightarrow x_0^2 + y_0^2 = 4y_0^2 \Rightarrow x_0^2 = 3y_0^2$.
From the hyperbola:
$\frac{3y_0^2}{4} - \frac{y_0^2}{8} = 1 \Rightarrow \frac{5y_0^2}{8} = 1 \Rightarrow y_0^2 = \frac{8}{5}$.
Area $= \frac{\sqrt{3}}{4}(2y_0)^2 = \sqrt{3} \cdot y_0^2 = \frac{8\sqrt{3}}{5}$.

5. **(4)** For ellipse $\frac{x^2}{36}+\frac{y^2}{16}=1$: $a=6$, $b=4$, $c=\sqrt{36-16}=2\sqrt{5}$.
Foci of ellipse: $(\pm 2\sqrt{5},0)$.
For hyperbola with eccentricity $e=5$ and foci at $(\pm 2\sqrt{5},0)$:
$e=\frac{C}{A}\Rightarrow A=\frac{2\sqrt{5}}{5}$.
$C^2=A^2+B^2\Rightarrow 20=\frac{4}{5}+B^2\Rightarrow B^2=\frac{96}{5}$.
Latus rectum $=\frac{2B^2}{A}=\frac{2\cdot\frac{96}{5}}{\frac{2\sqrt{5}}{5}}=\frac{96}{\sqrt{5}}$.

6. **(8)** $\frac{x^2}{8}-\frac{y^2}{8\cos^2\theta}=1$, $e_1=\sqrt{1+\frac{8\cos^2\theta}{8}}$

$$\ell_1=\frac{2b^2}{a}=\frac{2\cdot(8\cos^2\theta)}{2\sqrt{2}}$$

$$\frac{x^2}{6}+\frac{y^2}{6\cos^2\theta}=1;\ e_2=\sqrt{1-\frac{6\cos^2\theta}{6}}=\sin\theta$$

$$\ell_2=\frac{2b^2}{a}=\frac{2\cdot 6\cos^2\theta}{\sqrt{6}}$$

$$e_1^2=e_2^2(1+\sec^2\theta)$$

$$1+\cos^2\theta=\sin^2\theta\left(1+\frac{1}{\cos^2\theta}\right)$$

$$1+\cos^2\theta=\sin^2\theta+\tan^2\theta$$

Solving we get $\theta=\frac{\pi}{4}$

$$\ell_1=2\sqrt{2}$$

$$e_1=\sqrt{\frac{3}{2}}$$

$$\ell_2=\sqrt{6}$$

$$e_2=\frac{1}{\sqrt{2}}$$

$$\left(\frac{\ell_1\ell_2}{e_1e_2}\right)\tan^2\theta=8 \text{ (By putting values)}$$